

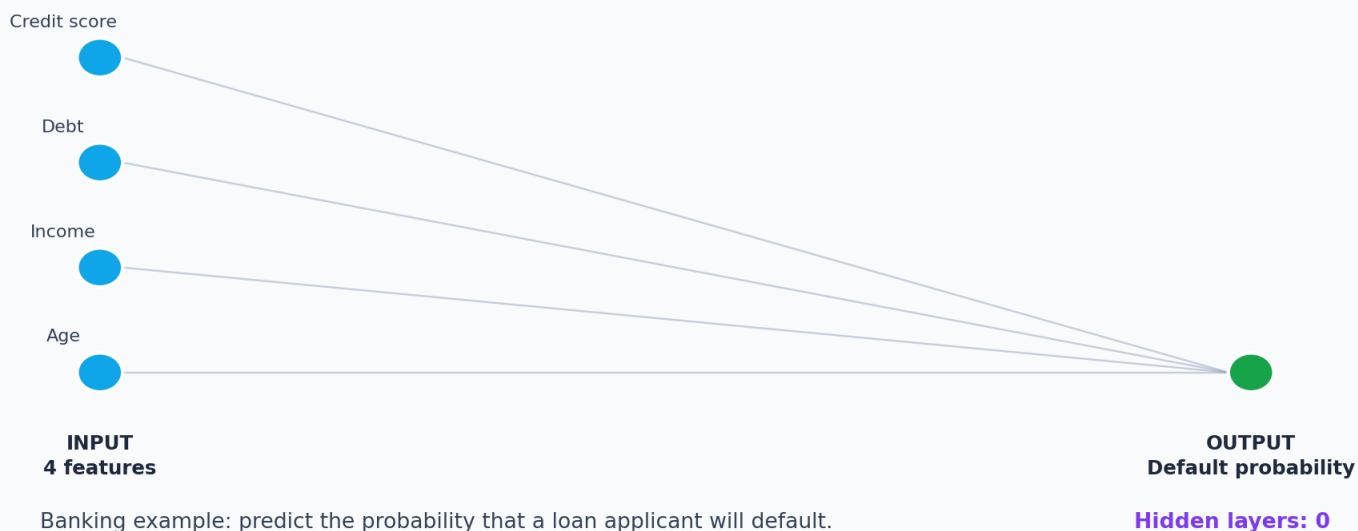
STEP 1 OF 8

Logistic regression: no hidden layer

STEP 1 OF 8 • FROM LOGISTIC REGRESSION TO ANN

Logistic regression

One weighted sum, then sigmoid. No hidden layer.



10 points to explain to participants

- 01.** The four blue circles are applicant features: age, income, debt, and credit score.
- 02.** Each input has a weight that controls its contribution to the prediction.
- 03.** The model multiplies each input by its weight and adds the results.
- 04.** It also adds a bias: an adjustable starting value.
- 05.** The resulting number is called a weighted sum.
- 06.** A sigmoid function turns that number into a value from 0 to 1.
- 07.** The green output shows the predicted probability of loan default.
- 08.** For example, 0.72 means a predicted 72% probability of default.
- 09.** There is no hidden layer between the input and the output.
- 10.** This is the starting point; later images add intermediate neurons.

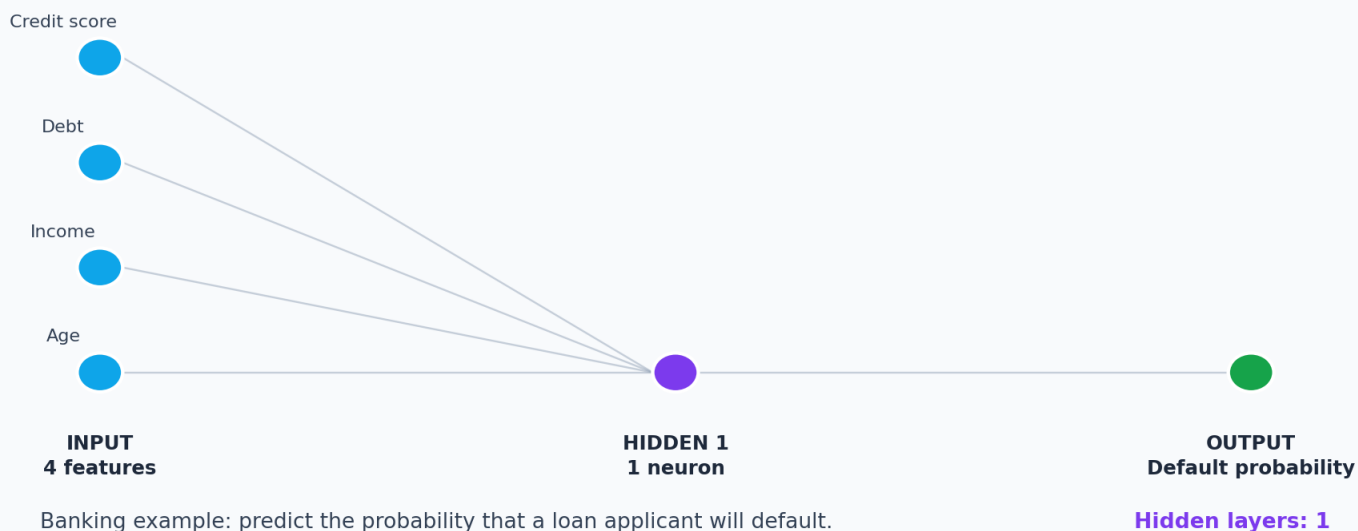
STEP 2 OF 8

ANN: one hidden neuron

STEP 2 OF 8 • FROM LOGISTIC REGRESSION TO ANN

ANN: one hidden neuron

One hidden neuron learns a simple intermediate pattern.



10 points to explain to participants

01. One purple hidden neuron now sits between the inputs and output.
02. Hidden means internal to the model, not a dataset column or final answer.
03. This neuron receives values from all four applicant features.
04. It computes its own weighted sum and adds a bias.
05. An activation function transforms that result.
06. The output neuron uses the hidden neuron result to make a prediction.
07. Think of it as learning one intermediate pattern in the data.
08. Training adjusts the weights and biases to improve predictions.
09. One hidden neuron has limited capacity to represent different patterns.
10. The new feature is an extra calculation between input and output.

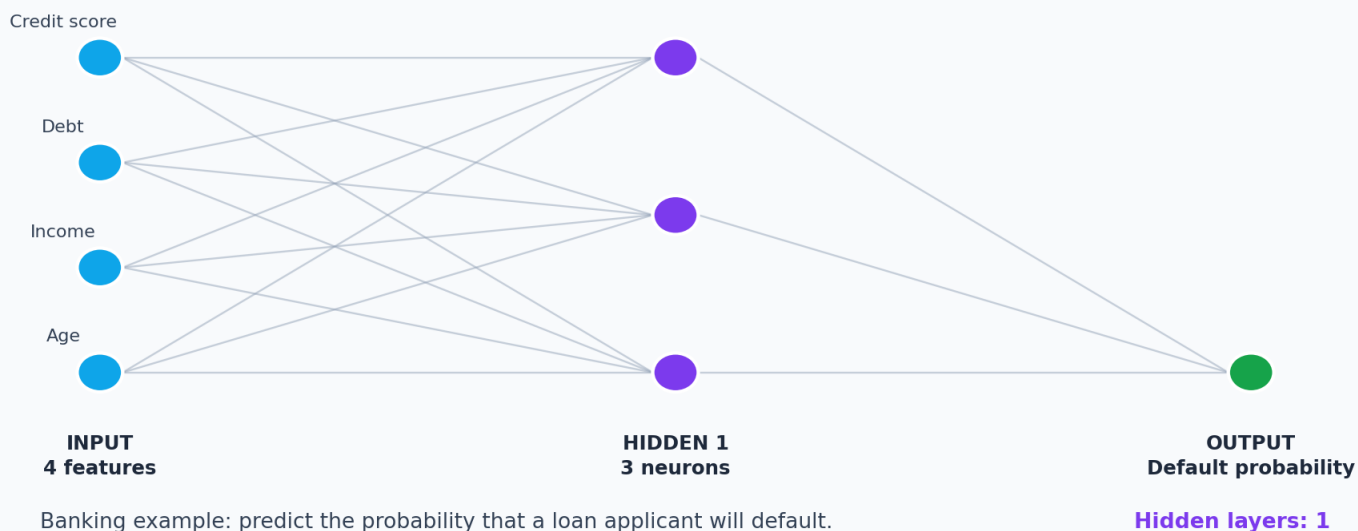
STEP 3 OF 8

ANN: one hidden layer, three neurons

STEP 3 OF 8 • FROM LOGISTIC REGRESSION TO ANN

ANN: one hidden layer

Three neurons learn different intermediate patterns.



10 points to explain to participants

01. There is still one hidden layer, but it contains three neurons.
02. Each hidden neuron receives the same four input features.
03. Each neuron has its own weights and bias.
04. They can respond differently to the same applicant.
05. One may respond to debt; another may respond to income and credit score.
06. These are possible patterns, not fixed meanings assigned to neurons.
07. The output neuron combines the results of the three hidden neurons.
08. More neurons in the same layer make a network wider.
09. It is not deeper because there is still only one hidden layer.
10. Several intermediate calculations now work in parallel.

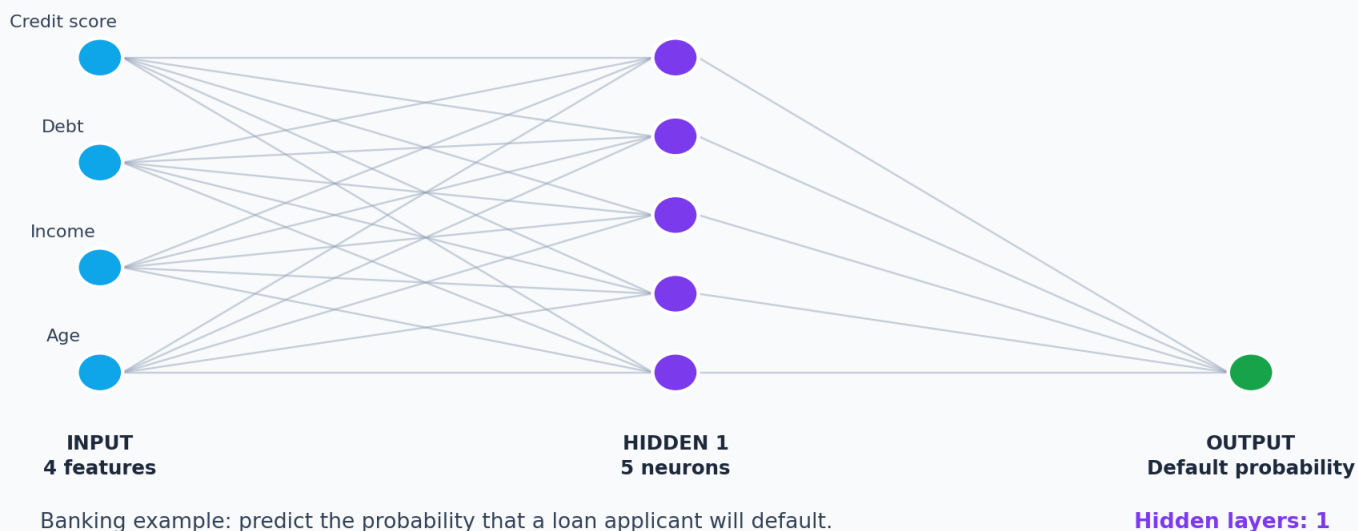
STEP 4 OF 8

Wider ANN: five neurons

STEP 4 OF 8 • FROM LOGISTIC REGRESSION TO ANN

Wider ANN

Five neurons increase the width of the same hidden layer.



10 points to explain to participants

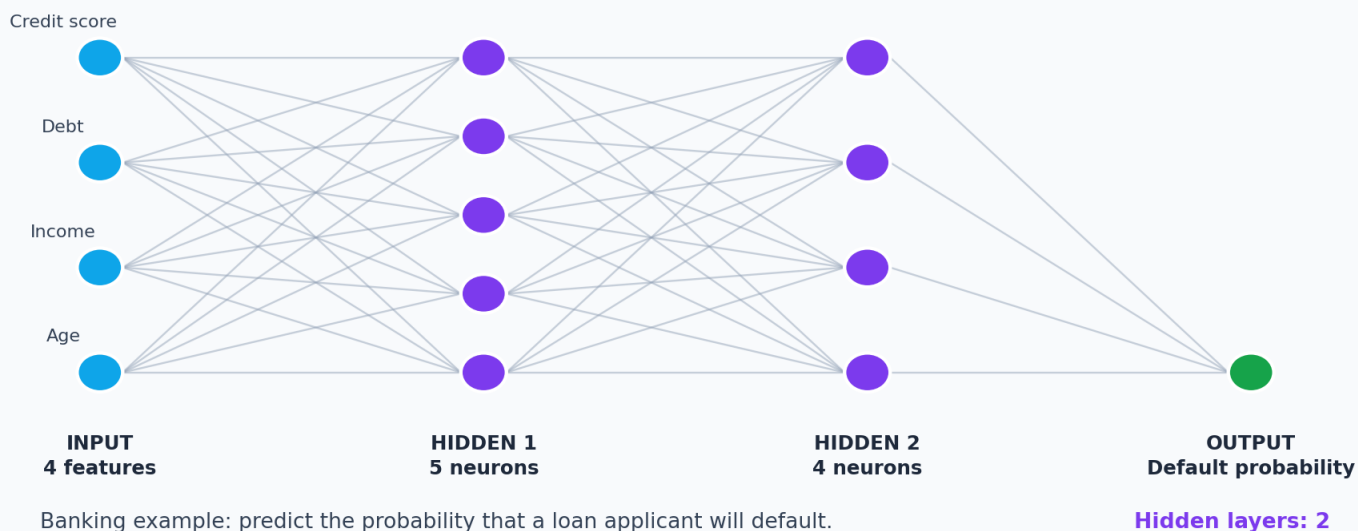
01. The hidden layer grows from three neurons to five.
02. The network is wider, but it still has one hidden layer.
03. Each neuron learns its own weights during training.
04. More neurons give the model capacity to represent more patterns.
05. The output neuron combines all five hidden neuron results.
06. Patterns may involve combinations of debt, income, and credit score.
07. More neurons also introduce more parameters to learn.
08. More capacity does not automatically produce better predictions.
09. Too much capacity for the data can cause overfitting.
10. Compare Images 3 and 4: width changes, while depth stays the same.

Deeper ANN: two hidden layers

STEP 5 OF 8 • FROM LOGISTIC REGRESSION TO ANN

Deeper ANN: two hidden layers

The second layer combines patterns learned by the first.



10 points to explain to participants

01. This is the first image with two hidden layers.
02. Hidden layer 1 receives the original four input features.
03. Hidden layer 2 receives the results of hidden layer 1.
04. The output neuron receives the results of hidden layer 2.
05. Information passes through two stages of intermediate calculations.
06. Nonlinear activations allow the network to learn nonlinear relationships.
07. The second layer can combine patterns found by the first.
08. Adding a hidden layer increases the network depth.
09. Training adjusts weights and biases across both hidden layers.
10. Compare with Image 4: a new layer was added, not just more neurons.

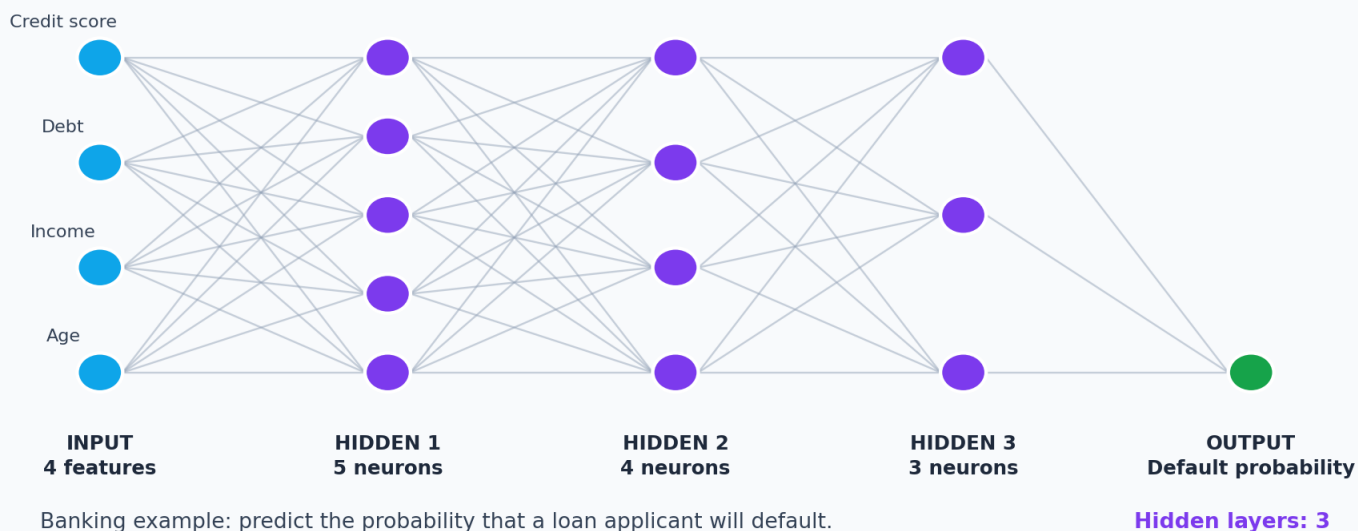
STEP 6 OF 8

Deeper ANN: three hidden layers

STEP 6 OF 8 • FROM LOGISTIC REGRESSION TO ANN

Deeper ANN: three hidden layers

A third transformation builds on the earlier layers.



10 points to explain to participants

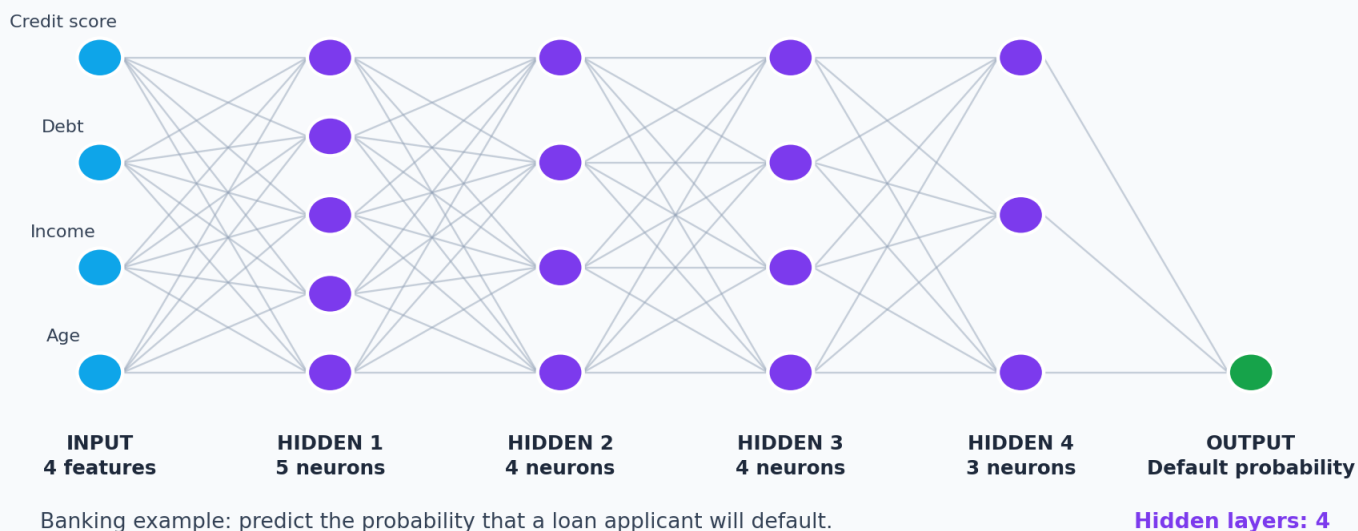
01. There are three hidden layers between the input and output.
02. Each hidden layer receives results from the preceding layer.
03. Each layer uses weights, biases, and activation functions.
04. Earlier layers can form intermediate representations of input data.
05. Later layers can combine those representations further.
06. The green output still means probability of loan default.
07. The prediction task is unchanged; only the model structure changes.
08. Deeper networks require more calculations and often more parameters.
09. Validation or test data tells us whether the extra depth helps.
10. Count purple columns, not purple circles, to count hidden layers.

Deeper ANN: four hidden layers

STEP 7 OF 8 • FROM LOGISTIC REGRESSION TO ANN

Deeper ANN: four hidden layers

Four hidden layers learn increasingly composed patterns.



10 points to explain to participants

01. This network has four hidden layers.
02. The four applicant features enter the first hidden layer.
03. Values pass through the hidden layers one after another.
04. Every connection has a weight learned during training.
05. Each hidden neuron also has a bias and an activation function.
06. Successive layers can represent increasingly combined patterns.
07. The final green circle remains one probability value.
08. Greater depth can make model training more demanding.
09. It may increase overfitting risk when the dataset is small.
10. Trace a path from an input through all four purple layers to output.

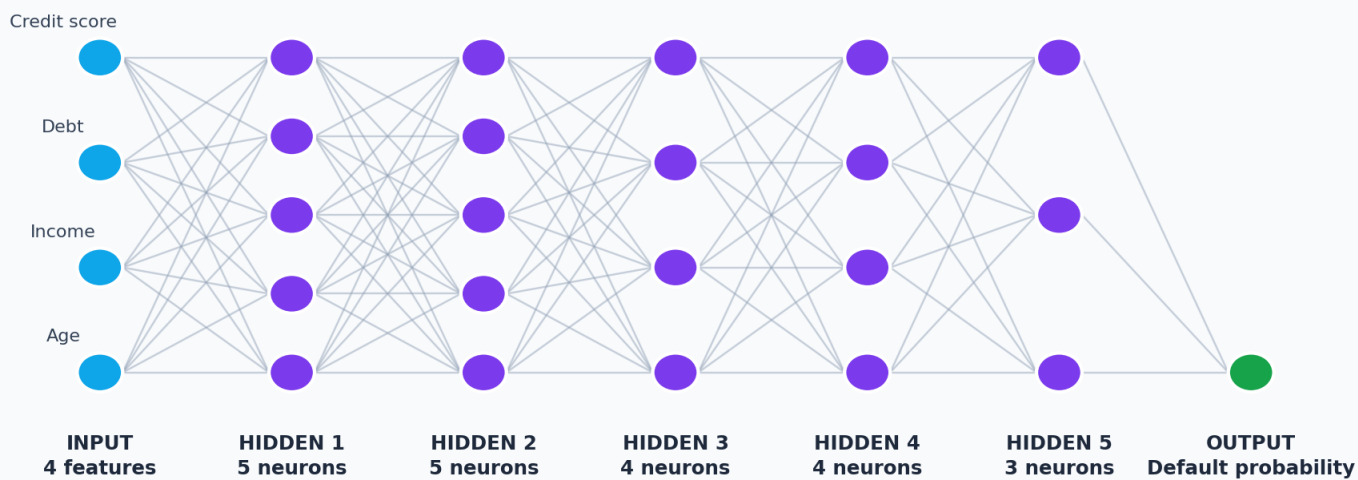
STEP 8 OF 8

Deep ANN: five hidden layers

STEP 8 OF 8 • FROM LOGISTIC REGRESSION TO ANN

Deep ANN: five hidden layers

Five hidden layers; depth counts layers, width counts neurons.



Banking example: predict the probability that a loan applicant will default.

Hidden layers: 5

10 points to explain to participants

01. This deepest network has five hidden layers.
02. Its inputs and prediction target are identical to Image 1.
03. The purple columns show five stages of internal processing.
04. Depth refers to the number of hidden layers.
05. Width refers to the number of neurons within a layer.
06. Each layer transforms values before passing them forward.
07. The final neuron produces a predicted loan default probability.
08. Training learns the weights and biases from example applicants.
09. Five layers do not guarantee higher accuracy than a smaller model.
10. Compare models on unseen data, training cost, and explainability.